Data availability

Chemical
pH (1:5)
E.C. (mS/cm)
Macronutrients (N,Ca, P...)
Micronutrients (Fe,Mb,Cu...)
Active Calcium
Carbonates (% CO_3)
CEC (meq/100g)
Heavy metals

Biological
Organic Matter (%)
C/N Ratio
Organic carbon (%)
Nematods genus
Fungi presence

Physical
Clay (%)
Silt (%)
Sand (%)
Texture

Risks

Operational risks (reversal after heatwaves, vandalism, contamination hotspots) are managed through zoning (high-risk areas capped or phytostabilised), portfolio buffers (a share of outcomes banked to cover setbacks) and seasonal rules that keep vulnerable surfaces covered during peak heat and traffic. Price and demand risks are lower than in farm carbon markets, but budget volatility is real; the scheme therefore diversifies payers (municipal departments, utility partners, CSR sponsors) and uses multi-year service agreements to smooth flows. Administrative burden is contained by the single-window operator, standard templates and batched audits; citizen-science inputs are curated, not relied upon for compliance.



Preliminary results

The expected outcomes are tangible in an urban park. **Soils gain SOM, lose compaction and hold water better**; runoff and heat pressures ease as cover improves and sealed areas shrink; **biodiversity returns** in ground flora and invertebrates where disturbance falls. Economically, the park unlocks new value streams tied to services the city already needs; stormwater retention, cooling, safer soils, housing revaluation; while the public base keeps routine care predictable. Institutionally, Tamarguillo Park demonstrates that a hybrid PES can translate urban soil stewardship into defensible contracts with credible, lightweight MRV, offering a template other city parks can adopt and scale

Economic perspective

The park's financing combines a basic operational layer with a results-based layer. The first ensures steady funding from the regular parks budget for routine maintenance and soil-friendly practices, providing cash flow stability. The second rewards measurable ecosystem outcomes, using transparent tariffs per unit delivered. Ecosystem-service valuation tools (e.g., i-Tree, InVEST, SWMM) translate these outcomes into monetary terms, strengthening accountability and co-funding opportunities.

Funding responsibility is shared: the base is covered by the parks budget, while results payments come from the departments that benefit directly, supplemented by CSR or climate funds. Multi-year agreements give financial certainty, and framework contracts plus digital monitoring reduce costs and administrative burden.

Overall, the model aligns financial flows with verified environmental services, providing predictable income for park operations while incentivising performance, efficiency, and scalability across districts.

Policy and governance

Governance is kept local and simple: the municipality leads contracts and oversight, while a programme operator manages results, data, and verification. Universities ensure scientific rigour, civil society contributes to monitoring, and planning/transport departments align works with ecological goals. Transparent MRV, based on soil tests, remote sensing and open dashboards, guarantees credibility at low cost.

Payment for Ecosystem Services in the Tamarguillo Park



novasoil

The NOVASOIL project aims to show the benefits for society and the environment of investing in soil health.

The main output of the project will be a tool to analyse the suitability of different business cases that promote soil health.

This tool will be based on best practice examples in Europe and other countries, as well as on the needs and demands of society.

Integrated Production

Summarize



The Tamarguillo Park case study project provides an opportunity to assess and improve the quality and health of the soil in Tamarguillo Park, which can have significant benefits for local biodiversity, water quality, drought resilience and the quality of life of local people. Healthy soils have good water-holding capacity, which is crucial for rainfall management and erosion control in the park.

The Tamarguillo Park case study project, a peri urban park, is an important initiative that will carry out soil sampling and soil health and quality assessments to identify the main features of a potential soil health business model, problems and opportunities, as well as the ecosystem benefits and revenues of the park, which will be communicated to key actors and stakeholders.

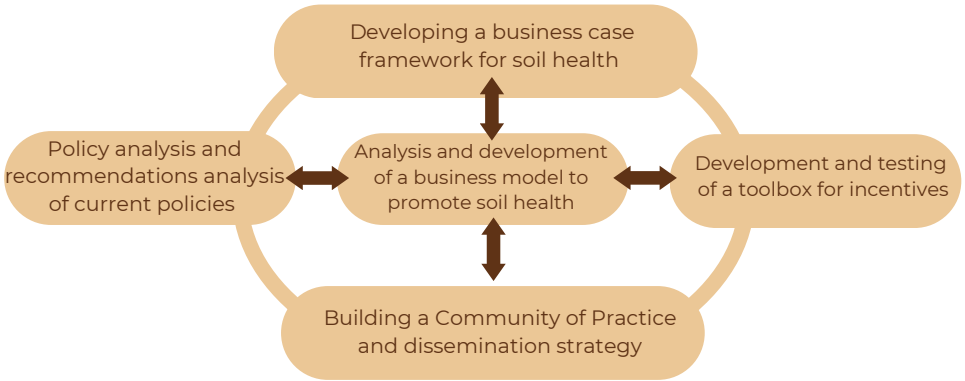


Figure 1. Structure of the NOVASOIL project

Case studies properties

Potential inhibitors

Economic crisis, climate change, lack of funding, lack of community involvement, conflicting interests, alignment with urban development plan.

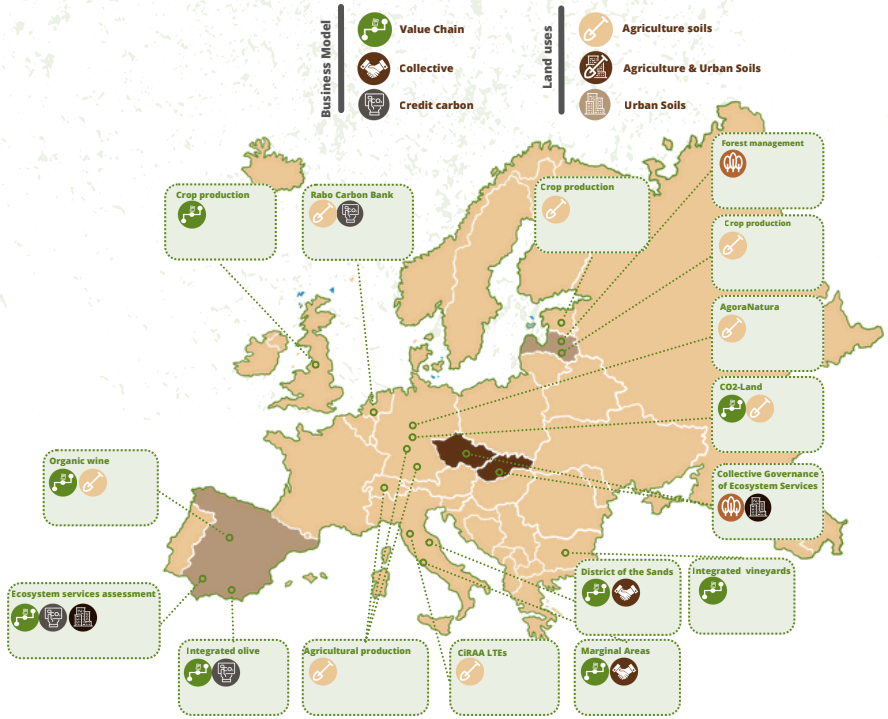
Potential enablers

Municipalities, communities, secure platforms, volunteers.

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Potential stakeholders

Local and regional authorities, non-governmental organisations (NGOs), local communities, voluntary groups, researchers and academics..



Ecosystem services



Specific objectives

SO1 Multi-dimensional portfolio development of business cases and good examples for investing in soil health in Europe and beyond.

SO2 Analysis of current models around the world for creating new incentives and new revenues from healthy soils.

SO3 Apply a co-development approach to modelling that promotes soil health.

SO4 Develop a prototype incentive toolbox and test network.

SO5 Analysis of current policies and technologies related to soil health and income provision for farmers.

SO6 Developing a community of practice around investors and key soil health actors.

SO7 Increase interest in soil health through a digital marketing strategy plan

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INNOVATIVE BUSINESS MODELS FOR SOIL HEALTH

https://novasoil-project.eu

